Reinforcement learning

Unit 5 · reference sheet · AI/ML in 35 topics

MDP

(S, A, P, R, γ); $\pi(a \mid s)$; $G_t = \sum_{k \geq 0} \gamma^k r_{t+k+1}$
Horizon $1/(1-\gamma)$: γ=0.9 $\Rightarrow$ 10, 0.99 $\Rightarrow$ 100

Values

$V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a)[R + \gamma V^\pi(s')]$
$Q^\pi(s,a) = \sum_{s'} P[R + \gamma V^\pi(s')] \quad V^* = \max_a Q^*$
TD: $V \leftarrow V + \alpha[r + \gamma V(s') - V(s)]$; bracket = TD error
Terminal state: $V = 0$, never bootstrap off it.

Exploration

ε-greedy: $P(\text{best}) = 1 - \varepsilon + \varepsilon/k$
UCB: $\arg\max_a \left[Q(a) + c\sqrt{\ln t / N(a)} \right]$
Regret $= T \cdot r^* - \sum_t r_t$; good algorithms $O(\log T)$

Two families

	Value	Policy
Learns	$Q(s,a)$	$\pi_\theta(a \mid s)$
Data	off-policy	on-policy
Actions	discrete	continuous OK
Failure	instability	variance

$Q \leftarrow Q + \alpha[r + \gamma \max_{a'} Q(s', a') - Q]$

Gotchas

- Reward is the spec: agents optimise exactly what you wrote.
- Greedy on noisy estimates locks onto the wrong arm forever.
- γ=0 is myopia, not efficiency.

This unit at a glance

#	Topic	You can
19	The RL frame	formalize a task as (S, A, P, R,) and define return
20	Value functions & the Bellman equation	write V and Q and explain the recursive expectation
21	Exploration vs exploitation	run -greedy by hand; explain regret on a bandit
22	Two solution families	contrast Q-learning (value) vs policy gradient (policy) on what they learn and

If you see this → do that

Signal	Move
No labels	RL or self-supervision
Reward is delayed	discount
Continuous actions	policy gradient
Discrete actions	DQN
Samples expensive	off-policy

Numbers worth memorising

γ=0.9: 10-step return = 6.51; horizon 10
γ=0.99: horizon 100; reward 11 at t=100 $\rightarrow$ 4.03
ε=0.1, k=4: best arm chosen 92.5%
TD example: r=1, γ=0.9, V'=0.8 $\Rightarrow$ Q=1.72